

**ANSWER TABLE****3.1 Factors influencing water availability**

| Scenario                                              | Key Factor(s)                                            | Explanation of Impact on Water Availability                                                                                              |
|-------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. A Dry Climate</b>                               | Climate (low rainfall, high evaporation)                 | Limited rainfall means little surface water; high temperatures increase evaporation, reducing both river flow and groundwater recharge.  |
| <b>2. Permeable Rock, Good Aquifers</b>               | Geology (permeable rock, strong infiltration)            | Water easily soaks into permeable rocks, creating reliable aquifers that provide steady groundwater supplies.                            |
| <b>3. Steep Mountain Slopes</b>                       | Topography (steep slopes)                                | Water runs off quickly, reducing infiltration and creating short-lived streams that don't store water well.                              |
| <b>4. Deforested Catchment</b>                        | Vegetation loss, land use                                | Less vegetation increases runoff and erosion, lowering infiltration and groundwater recharge and reducing water quality.                 |
| <b>5. Expanding City</b>                              | Human demand, infrastructure pressure                    | High water use from population and industry puts pressure on dams and supply systems, causing shortages even in moderate rainfall areas. |
| <b>6. Ageing Infrastructure in a Remote Community</b> | Infrastructure, remoteness                               | Old or unreliable equipment limits access to safe drinking water, even when water sources exist. Repairs take longer due to remoteness.  |
| <b>7. Desert Basin (Inland Drainage)</b>              | Climate, natural drainage patterns, topography           | Water collects in internal basins and evaporates quickly in hot conditions; little long-term storage and often saline water.             |
| <b>8. Dam Controlling River Flow</b>                  | Human modification of natural systems (dams, management) | Upstream dams reduce downstream river flow, limiting natural water availability for communities and ecosystems.                          |
| <b>9. Fertiliser Runoff</b>                           | Land use, pollution                                      | Nutrient runoff causes algal blooms and lowers water quality, making water difficult or unsafe to use without treatment.                 |
| <b>10. Island Community</b>                           | Limited storage, climate, catchment size                 | Small islands rely on rainfall; limited groundwater and small catchments mean supplies run out quickly in dry seasons.                   |

## **ANSWER SHEET 5.1 Top causes of water scarcity**

### **1. Climate Change**

Definition: Long-term changes in temperature and weather patterns that reduce rainfall or increase evaporation, leading to less available water

Example: A region becomes hotter and receives less rain, causing rivers and reservoirs to shrink.

### **2. Population Growth**

Definition: An increase in the number of people using water, which raises demand beyond what local water supplies can provide.

Example: A fast-growing city struggles to supply enough drinking water to its expanding population.

### **3. Agricultural Demand**

Definition: Large amounts of water are needed for crops and livestock; inefficient irrigation or farming practices can use more water than the environment can replace.

Example: Farmers use open-channel irrigation that loses water to evaporation, reducing the amount left in local rivers.

### **4. Pollution of Freshwater Sources**

Definition: Contamination of rivers, lakes, or groundwater by chemicals, waste, or harmful substances that make the water unsafe to use.

Example: A factory dumps waste into a river, making the water unsafe for drinking or irrigation.

### **5. Over-extraction of Groundwater**

Definition: Removing groundwater from underground aquifers faster than it can naturally refill.

Example: Wells in a farming region run dry because too much water was pumped for irrigation.

### **6. Poor Water Management Infrastructure**

Definition: Old, damaged, or poorly planned water systems that waste water through leaks or fail to deliver it effectively.

Example: A city loses large amounts of water through leaking pipes before it ever reaches homes.

### **7. Economic and Political Factors**

Definition: Lack of funding, weak government policies, conflict, or inequality that prevents people from accessing available water.

Example: A rural village has a nearby river but no investment in pipes or pumps to deliver the water.

### **8. Urbanisation**

Definition: Rapid growth of cities that increases water demand and puts pressure on existing water supplies and infrastructure.

Example: A growing metropolis uses more water than its reservoirs can supply, causing seasonal shortages.

**Short: term strategies**

1. **Water restrictions and rationing:** Limiting how much water households, farms, and businesses can use to conserve water immediately.
2. **Public awareness campaigns:** Teaching people how to save water during shortages.
3. **Installing water: saving devices:** Using taps, showerheads, or toilets that use less water to reduce consumption quickly.
4. **Emergency tanker water supply:** Bringing extra water from outside areas when local sources are too low.

**Long: term strategies**

5. **Building a dam:** Storing large amounts of water to provide a reliable supply over many years.
6. **Improving irrigation and piping:** Updating irrigation systems and pipelines to reduce water loss over time.
7. **Rainwater harvesting:** Collecting rainwater from roofs or surfaces for ongoing use.
8. **Desalination plant:** Turning seawater into drinking water to create a long: term water source.
9. **Improving irrigation efficiency:** Using smarter irrigation methods (like drip systems) to save water in farming.
10. **Recycling wastewater:** Treating used water so it can be safely reused for irrigation, industry, or even drinking.